

Year 8 Science – Booklet 20: Chemistry

Chemical Reactions — ANSWERS

Spot Check (p.61)

1. Exothermic.
2. Magnesium + oxygen → magnesium oxide; $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
3. Combustion.

Chemical reactions (workbook, p.154–155)

1. Reversibility of changes

- a). A physical change.
- b). E.g. melting ice, freezing water, or boiling water into steam.
- c). E.g. burning wood/fuel, rusting of iron, cooking an egg, or baking a cake.

2. Iron and oxygen

- a). Combustion.
- b). Iron oxide.
- c). Rust.

3. Sodium + chlorine → sodium chloride

- a). sodium + chlorine → sodium chloride
- b). $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$

4. Sodium chloride

- a). Salt (table salt).
- b) i). Sodium is a soft, reactive, silvery metal.
- b) ii). Chlorine is a poisonous, greenish-yellow gas (a non-metal, halogen).

5. Fire and a browning apple – same reaction

Oxygen.

6. Chemical reaction that takes in heat

Endothermic.

7. Respiration is a chemical reaction

- a). Glucose and oxygen.
- b). Carbon dioxide, water (and energy).